

BUSINESS + TECHNICAL SUMMARY

WISER DOM Hybrid Solver

A portable classical default with safeguarded quantum experimentation

Executive decision

Adopt **polished greedy** as the portable planning default and use **exact large-neighborhood search (LNS)** when a planning window contains valuable inventory or dock conflicts. Keep the sampler-assisted hybrid, including quantum hardware, as a controlled research option behind exact recourse and independent validation.

On the common 20-assignment-group subset, greedy increased objective capture from 61.39% to 64.90% and case fill from 64.21% to 68.50%. Exact polishing retained that solution in 2.86 seconds. Exact LNS, full exact mixed-integer linear programming (MILP), and the simulated-annealing hybrid reached the same displayed quality. The full exact solver certified this small subset with zero optimality gap.

Business problem

Distributed order management (DOM) selects a distribution center (DC) and planned ship date while orders compete for shared inventory and capacity. Orders sharing a load must move together, a non-default DC must improve service, and no recommendation may exceed cumulative available-to-promise (ATP) inventory or enabled capacity.

The optimized business objective is:

fulfilled merchandise value – unmet-demand penalty – shipping cost.

Objective capture divides that result by total requested merchandise value, making comparisons interpretable without disclosing commercial totals. **Case fill** is fulfilled cases divided by requested cases. **Reassignment** means moving an order away from its default DC.

Solution hierarchy

Layer	What it does	Recommended use
Default baseline	Keeps planned routing and allocates feasible quantities	Business comparison
Greedy	Chooses whole-load alternatives while updating shared resources	Fast initial plan
Polished greedy	Fixes routing and exactly re-optimizes quantities and penalties	Routine production default
Exact LNS	Re-optimizes bounded groups of interacting decisions	Quality escalation
Full exact MILP	Solves all selected decisions and reports a bound	Small certificates and benchmarking
Hybrid sampler	Proposes coordinated local assignments; exact models retain control	Research only

The hybrid's quantum unconstrained binary optimization (QUBO) covers only a bounded local neighborhood. QAOA, simulated annealing, random sampling, or exact enumeration can propose binary choices. The system repairs one-hot violations, exactly re-optimizes fulfillment, validates the full plan, and rejects every invalid or non-improving result.

Evidence

The reviewed study contains 247 aggregate rows across 14 experiment families. All 247 passed the independent validator. Maximum numeric residuals for demand balance, integrality, cumulative inventory, and enabled capacities were zero.

Common-subset method	Objective capture	Case fill	Runtime
Default routing	61.39%	64.21%	0.30 s
Greedy	64.90%	68.50%	0.76 s
Polished greedy	64.90%	68.50%	2.86 s
Exact LNS	64.90%	68.50%	7.00 s
Full exact MILP	64.90%	68.50%	2.76 s
Hybrid simulated annealing	64.90%	68.50%	12.88 s

Scaling reached 372 assignment groups and 750 orders. Median runtime there was 16.74 seconds for greedy, 54.63 for polished greedy, and 74.03 for exact LNS. LNS produced small additional gains at several larger scales, supporting time-budgeted escalation.

A synthetic greedy trap was improved by 197.2 units by exact LNS, exact MILP, simulated annealing, and local QAOA, increasing case fill from 21.49% to 30.58%. This validates coordinated neighborhoods, not quantum advantage; exact classical methods were faster.

Under a 40% aggregate inventory shock, re-optimized routing achieved 69.66% case fill versus 67.21% with the original routing fixed. This is the clearest operational robustness result: the solver can respond to changed availability instead of only reallocating quantities along stale routes.

Quantum assessment

The IBM archive contains 18 QPU jobs on a generated 16-qubit control. All QPU-derived plans were feasible after exact recourse, but native one-hot feasibility fell from approximately 51%–59% at $p=1$ to 12%–15% at $p=2$ as median two-qubit gates rose from 122 to 420. The QPU did not beat exact MILP in quality or time.

The final code strengthens the next experiment with a lower-edge path mixer, linear W-state preparation, reusable optimized angles, confidence intervals, near-optimal rates, and uniform-feasible controls. Those circuit changes require a matched hardware rerun; they are not retroactively credited to the archived IBM result.

Deployment and next decisions

The default stack is license-free and runs on Windows, macOS, and Linux with Python 3.10–3.12. SciPy/HiGHS is the default exact backend. Gurobi is a worthwhile optional comparison because it may improve solve progress on larger MILPs, but it is neither a dependency nor the default; the notebook skips it cleanly without a package or license.

Before a pilot, owners should confirm authorized alternative DCs, the working-day calendar, throughput limits, customer rules, and approval thresholds. Run polished greedy on rolling historical windows; trigger LNS only for material conflicts; review validator diagnostics and planner acceptance; and compare realized service with default routing. Restrict quantum reruns to approved synthetic or bounded neighborhoods with exact and uniform-feasible controls.

The result is a usable classical solver and a safe quantum test bed whose operational quality does not depend on a license, GPU, or quantum account.